

SOAEB

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SUMMARY

3rd year CSE student building full-stack and ML projects. Experienced in Python, Django, and Machine Learning. Open source contributor (Kolibri). Mentor at GSSOC. Looking for backend/ML engineering internship where I can solve real problems and learn from experienced teams.

EXPERIENCE

Open Source Contributor | Kolibri (Learning Equality) (Jun'26 - Present)

- Migrating and testing Vue component test suite from @vue/test-utils to Vue Testing Library
- Designed 8+ comprehensive test cases covering user interactions, async operations, and error handling
- All tests passing with code review approval from maintainers
- Learning: VTL best practices, test-driven development, Git workflow

Machine Learning Internship | IIT Roorkee (Feb '26 - April '26)

- Developed multi-view transformer model (MalBERT-XAI) for Android malware detection using PyTorch
- Achieved 99.5% binary accuracy and 94.3% family classification across 5 malware families
- Built full ML pipeline: APK parsing → feature extraction (permissions, API calls, opcodes, intent) → tokenization → model training → evaluation
- Implemented interpretability layer using LIME to explain model predictions
- Key Learning: Transformer architectures, multi-task learning, model interpretability

PROJECTS

AI-Powered Organ Matching Platform (Django | Python | Scikit-learn | Tailwind | HTMX) ([Live Project](#))

Problem: Manual donor-recipient matching is slow and error-prone in organ transplant systems. achieving 95% model accuracy on a dataset of 900+ records

Solution Built:

- Designed ML pipeline matching donors to recipients using TF-IDF + cosine similarity on 15+ clinical features (blood type, urgency, proximity, etc.) — 95% accuracy on 900+ records
- Full-stack: Backend REST API (Django) + Frontend (Tailwind/HTMX) + ML model serialization
- Implemented business rules layer (blood type constraints, urgency prioritization) on top of ML scores
- Achieved sub-second inference time for real-time recommendations in production environment

Key Learning: Balancing ML accuracy with domain-specific business rules

BarberBook — Full-Stack Mobile Booking Marketplace ([Git Repo](#)) ([Live App](#))

- **Problem:** Local salons rely on phone calls/walk-ins, causing double-bookings, no-shows, and fragmented staff or schedule management.
- **Solution:** Built an end-to-end, two-sided mobile app and REST API backend with distinct customer/owner experiences. Integrated secure OTP auth, stateless JWT sessions, HMAC-verified Razorpay payments, time-slot logic, and token-scoped Firebase push notifications across 10+ screens via a global theme file.
- **Key Learning:** Architecting for JWT statelessness and single-source-of-truth theming early drastically reduced the cost and complexity of future security and visual changes.

Personal Portfolio Website | (Python | Django | Tailwind | ORM) ([Live Portfolio](#))

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript

Backend & Databases: Django, Django REST Framework, Flask, SQLite, MySQL, Django ORM, REST APIs

ML & Data: Scikit-learn, Pandas, NumPy, PyTorch, Matplotlib, Seaborn

Frontend: React.js, Vue.js, HTML5, CSS3, Tailwind, HTMX

Tools & DevOps: Git, GitHub, Postman, VS Code, Jupyter Notebook, Linux CLI

ACHIEVEMENTS & RECOGNITION

- Open Source Contributor: Successfully migrated Vue component test suite with 100% passing tests and maintainer approval (Kolibri project, Learning Equality)
- Mentor: Guided 100+ students in full-stack development, Machine Learning.

EDUCATION

B.Tech. Computer Science & Engineering | Phonics Group of Institution, Roorkee (Graduating 2027)

- **Relevant Coursework:** Data Structures, Algorithms, Database Management, Web Development, Artificial Intelligence, Operating Systems, Computer Networks etc.

XII | Atal Utkrisht Government Inter College, Roorkee (2022) X |

Cantt Board Sr. Sec. School, Roorkee (2020)